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Data Analytics Internship

Understanding the HIMS Dataset

(Hospital Information Management System)

Project Step: Data Understanding (Before Data Cleaning & Analysis)

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Introduction

HIMS stands for Hospital Information Management System. The Excel workbook analyzed here is not a single spreadsheet — it is a relational database made up of 19 connected tables. Each table stores one part of hospital operations, and together they describe the complete journey of a patient through the hospital.

This document explains, in simple human language, what each table represents, what fields it contains, how many records it holds, and what real-world question it helps answer. This understanding is the necessary first step before Data Quality Checking, Cleaning, and further Analysis in the internship project.

The Patient Journey — In One Story

A patient comes to the hospital → gets admitted → is assigned a department, ward and bed → doctors examine the patient → diagnostic tests may be performed → medicines are prescribed → a bill is generated → insurance may cover part of the bill.

The HIMS dataset records this entire journey, end to end.

1. Patient & Admission — The Core Story

1. Patient

Table name: *patient* | **Records:** 30,000

Fields it stores:

- Patient ID
- Gender
- Date of birth
- Blood group
- City
- Contact number

This table answers:

Who is the patient, and what are their basic details?

2. Admission

Table name: *admission* | **Records:** 45,000

Fields it stores:

- Admission ID
- Admission date
- Discharge date
- Admission type

- Admission status
- Patient
- Department
- Ward
- Bed
- Disease

This table answers:

When did the patient come to the hospital, why were they admitted, where were they treated, and when were they discharged?

Example:

- Patient 166 → Emergency admission → Internal Medicine department → Ward 6 → Bed 76 → Disease ID 10
- Admitted Feb 25, 2020 → Discharged Feb 27, 2020

3. Disease

Table name: *disease* | **Records:** 20

Fields it stores:

- Disease ID
- Disease name
- Disease category

Human question this table answers:

What diseases are patients being treated for?

Example:

- Acute Myocardial Infarction → Cardiac
- Stroke → Neurological

2. Hospital Infrastructure

4. Department

Table name: *department* | **Records:** 11

Fields it stores:

- Department name
- Department type
- Floor number
- Status

This table answers:

| *Which department is responsible for the patient's treatment?*

Example:

- Emergency, Internal Medicine, Surgery, Pediatrics, Orthopedics, etc.

5. Ward

Table name: *ward* | **Records:** 27

Fields it stores:

- Ward name
- Ward type
- Total beds
- Department

This table answers:

| *Where inside the department is the patient staying?*

Example:

- Emergency Department → Emergency Ward 1, Emergency Ward 2, Emergency Ward 3 ...

6. Bed

Table name: *bed* | **Records:** 415

Fields it stores:

- Bed ID
- Bed number
- Bed status
- Ward

This table answers:

| *Which specific bed is being used, and is it occupied?*

Example:

- Bed status can be Occupied / Available / etc.

3. Doctors & Staff

7. Doctor

Table name: *doctor* | **Records:** 98

Fields it stores:

- Doctor ID
- Employee ID
- Specialization
- Qualification
- Experience (years)

This table answers:

| Which doctor is available, and what is their specialization and experience?

8. Employee

Table name: *employee* | **Records:** 500

Fields it stores:

- Employee name
- Gender
- Role
- Employment type
- Date of joining
- Department

This table answers:

| Who works in the hospital, and what role do they perform?

Example:

- Roles include Doctor, Nurse, Admin, and other hospital staff.

9. Staff Assignment

Table name: *staff_assignment* | **Records:** 207

Fields it stores:

- Employee
- Ward
- Shift

Human question this table answers:

| Who is working in which ward, and during which shift?

Example:

- Employee 2 → Ward 24 → Morning shift

4. Medicines

10. Drug

Table name: *drug* | **Records:** 250

Fields it stores:

- Drug name
- Brand name
- Drug category
- Unit cost
- Manufacturer

This table answers:

| *What medicines does the hospital use, and how much does each medicine cost?*

11. Drug Inventory

Table name: *drug_inventory* | **Records:** 250

Fields it stores:

- Current stock
- Reorder level
- Inventory status
- Last restock date
- Drug

This table answers:

| *Do we have enough medicine in stock, or do we need to reorder?*

Example:

- Useful for a dedicated inventory analysis project.

12. Prescription

Table name: *prescription* | **Records:** 73,109

Fields it stores:

- Dosage
- Frequency
- Duration (days)
- Admission
- Drug

This table answers:

| *What medicine was prescribed, how much, how often, and for how many days?*

Example:

- Example: Admission → Drug → 1 tablet → Twice a day → 12 days

5. Medical Tests

13. Diagnostic Test

Table name: *diagnostic_test* | **Records:** 9

Fields it stores:

- Test name
- Test category
- Standard cost
- Department

Human question this table answers:

| *What medical tests does the hospital perform, and what is their standard cost?*

Example:

- X-Ray Chest, CT Scan Brain, MRI Spine, Ultrasound Abdomen, Complete Blood Count, etc.

14. Patient Diagnostic

Table name: *patient_diagnostic* | **Records:** 63,269

Fields it stores:

- Test date
- Result status
- Admission
- Test
- Doctor

This table answers:

| *Which tests were performed for a patient, and what were the results?*

6. Billing

15. Billing

Table name: *billing* | **Records:** 45,000

Fields it stores:

- Bill date
- Total amount
- Insurance-covered amount
- Patient payable amount
- Payment status
- Payment mode
- Admission

This table answers:

How much did the treatment cost, how much did insurance pay, and how much did the patient have to pay?

Example:

- Total bill = ₹68,483
- Insurance = ₹61,634.70
- Patient pays = ₹6,848.30

16. Billing Details

Table name: *billing_details* | **Records:** 112,402

Fields it stores:

- Charge type (Room, Drug, Test, etc.)
- Reference ID
- Amount
- Bill

This table answers:

What exactly did the hospital charge the patient for?

Example:

- Instead of just 'Patient's bill = ₹68,483', this breaks it down:
- Room = ₹4,884 | Drugs = ₹1,844 | Other charges = ...

7. Insurance

17. Insurance Provider

Table name: *insurance_provider* | **Records:** 50

Fields it stores:

- Provider name
- Provider type

- Contact details
- Coverage limit

This table answers:

Which insurance companies are associated with the hospital, and what coverage do they provide?

18. Patient Insurance

Table name: *patient_insurance* | **Records:** 21,617

Fields it stores:

- Policy number
- Coverage percentage
- Policy start date
- Policy end date
- Patient
- Insurance provider

Human question this table answers:

Does this patient have insurance, which provider do they use, and how much of their treatment can be covered?

8. Medicine Manufacturers

19. Drug Manufacturer

Table name: *drug_manufacturer* | **Records:** 300

Fields it stores:

- Manufacturer name
- Country
- Reliability rating
- Contract status

Human question this table answers:

Who supplies our medicines, where are they from, and how reliable are they?

9. How the Tables Connect

Understanding these relationships is essential before starting SQL queries or Power BI dashboards. The central table connecting most of the hospital story is Admission — it links the patient to the department, ward, bed, disease, prescriptions, tests and billing.

10. Questions This Dataset Can Answer

Patient Analysis

| *Which patients are visiting the hospital most?*

Hospital Analysis

| *Which department has the most admissions?*

Bed Analysis

| *Which wards have the highest bed occupancy?*

Medicine Analysis

| *Which medicines are prescribed most frequently?*

Diagnostic Analysis

| *Which tests are performed most often?*

Financial Analysis

| *Which departments generate the highest billing?*

Insurance Analysis

| *How much of hospital revenue comes from insurance?*

Doctor Analysis

| *Which doctors handle the most admissions/tests?*